VaaniShield

Code-Switch-Aware Deepfake Voice Detection for Financial Fraud Calls in Indian Multilingual Speech

AI Project Submission — Abstract & Problem Statement
Prepared by: Dhruv | B.Tech CSE (AI), Manipal Institute of Technology, Bengaluru | September 2026

Abstract

Picture this: your phone rings. It's your father's voice — panicked, urgent, telling you he's in trouble and needs money sent right now. Except it isn't him. It's a machine that stole a few seconds of his voice from an old video call or a social media clip, and used it to build a near-perfect copy. This isn't science fiction — it's already happening across India, and it's spreading fast.

A lot of smart people are already trying to catch these fake voices. But almost every tool built so far was trained to listen to people speaking one language at a time — clean English, or clean Hindi, start to finish. Real Indian conversations don't work that way. We jump between English and our mother tongue mid-sentence, the way we always have. And that's exactly where the fake voices give themselves away: right at the moment a real speaker switches languages, a cloned voice tends to stumble — a tiny, almost invisible glitch that a machine trained on "clean" speech was never taught to listen for.

VaaniShield is built to catch exactly that stumble. It listens for the moment a caller switches languages mid-sentence, checks whether that switch sounds naturally human or subtly "off," and flags the call as suspicious — without needing to already know who the caller is. Think of it as a lie detector tuned specifically to the way Indians actually talk, built to catch a threat that every existing tool has quietly overlooked.

Problem Statement

India's high smartphone/UPI penetration, trust in phone-based banking, and linguistic diversity make it an especially exposed market for voice-cloning fraud, but the obvious response — an Indian-accent audio+text fusion detector shipped as a real-time phone tool — is already established territory: IndieFake/InDeepFake and a published CNN-Transformer model cover Indian-accent audio deepfake detection; Korean-vishing fusion systems and RoboKA already fuse "what is said" with "how it is said"; and SentinelEdge already implements real-time, on-device, federated scam-call detection in open-source code. What none of this addresses is linguistic structure: real Indian phone conversation, especially the urgent register fraud calls imitate (e.g. "beta, emergency hai, turant paise transfer karo"), routinely code-switches mid-sentence, and voice-cloning systems – trained on monolingual or cleanly-separated multilingual data – are not explicitly optimized to model the prosodic transition at such switch points. Separately, existing partial-spoof research locates *edited* splice boundaries, and a recent finding (SNAP) shows self-supervised speech encoders suffer "speaker entanglement" that hurts generalization to unseen callers. Neither thread has been connected to natural code-switch points in code-mixed Indian speech — a gap with direct fraud relevance, since this is precisely the register scam calls exploit.